

Single-molecule PCR as a branching process: reproducing the quPCR Fig. S3 and bounding the per-cycle AE noise

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Abstract

We simulate the single-molecule Cq distribution of quantitative PCR as a Galton–Watson process with per-cycle duplication probability AE, first with a literal reading of the quPCR supplementary Fig. S3 and then with the authors’ own code. The authors’ protocol reproduces both published panels (CqP = 38.22 and 38.32 against 38.21 and 38.31). The four-peak structure is a frozen record of the first three cycles; per-cycle efficiency noise smears it diffusively with width $s_\eta \simeq \sigma\sqrt{n}/[(1 + \text{AE}) \ln(1 + \text{AE})]$, so Fig. S3B tests a single noise level and sets an upper bound rather than excluding fluctuations. At small σ the four peaks return: peak 3 survives for $\sigma \lesssim 0.006\text{--}0.011$ and peak 4 for $\sigma \lesssim 0.04$ (AE = 0.959). A likelihood fit to the authors’ 297 single-copy wells gives $\sigma \lesssim 0.02\text{--}0.036$ (95%), robust to the assumed measurement noise. At AE = 0.8 the early-failure probability is five times larger and the side peaks are already merged at $\sigma = 0$; there are no peaks to return to, and bounding σ takes 10–30 $\times$ more wells.

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1 Model and notation

Each run starts from one single-stranded template, $N_0 = 1$. In cycle n every molecule independently duplicates with probability p_n ,

$$\begin{aligned} N_n &= N_{n-1} + \text{Bin}(N_{n-1}, p_n), \\ p_n &= \text{AE} + \sigma \xi_n, \quad \xi_n \sim \mathcal{N}(0, 1) \text{ i.i.d. in } n, \end{aligned} \tag{1}$$

with one draw of p_n per cycle and per run, shared by all molecules of that run. When p_n leaves $[0, 1]$ it is clipped (the authors' code does this implicitly: a molecule replicates if a uniform number is $\leq p_n$). Cq is the cycle at which N_n first reaches a threshold T , with log-linear interpolation between cycles. We use two readings of Fig. S3.

Definition 1.1 (Literal reading). $T = 2^{38}$, the recursion (1) runs until threshold, and “AE = 0.959 ± 0.1 ” means $\sigma = 0.1$ in every cycle.

Definition 1.2 (Authors' protocol). From the authors' MATLAB scripts for Fig. 1D and Fig. S3B (<https://github.com/Azuresky99/quPCR>):

1. “ ± 0.1 ” is the half-width of a 95% interval, $\sigma = 0.1/1.96 = 0.051$, with mean AE = 0.9593. (YS: the supplement text says mean 0.95; the code uses 0.9593)
2. Only 19 cycles are stochastic. Afterwards $\text{Cq} = \log(T/N_{20}) / \log(1 + \text{AE}) + 19$ at fixed AE, so cycles 21–38 carry no noise.
3. $T = (1 + \text{AE})^{\text{quCq}}$ with $\text{quCq} = 38.34$, tuned by hand until the simulated mode equals the measured $\text{CqP} = 38.21$. This is $T \approx 2^{37.2}$.

All runs use only the Julia standard library: the binomial is sampled exactly for $N \leq 2^{16}$ (geometric jumps over the rarer outcome) and by a Gaussian above.

2 Why there are peaks

Once N is large the binomial noise is suppressed as $N^{-1/2}$ and growth is deterministic, $N_n \simeq W(1 + \text{AE})^n$ with W the martingale limit of the Galton–Watson process. Hence

$$\text{Cq} \simeq \log_{1+\text{AE}} T - \log_{1+\text{AE}} W, \tag{2}$$

and the shape of the Cq distribution is the law of $\log W$, which is fixed during the first few cycles. Its sharp features are single missed duplications.

Peak 4 is exactly one cycle late. If the template fails in cycle 1, the process restarts one cycle later, so the density obeys the renewal relation

$$P(\text{Cq}) = \text{AE} P(\text{Cq} \mid N_1 = 2) + (1 - \text{AE}) P(\text{Cq} - 1). \tag{3}$$

Peak 4 is the main peak shifted by exactly +1, with weight $1 - \text{AE} = 4.1\%$; the simulations give +1.000 in both readings.

Peaks 2 and 3. One failure among the 2^{k-1} molecules of cycle k lowers W by the factor $(2^k - 1)/2^k$, i.e. a delay

$$\Delta_k \approx \frac{\ln [2^k / (2^k - 1)]}{\ln(1 + \text{AE})}, \quad (4)$$

which gives $\Delta_3 \approx 0.20$ (peak 2) and $\Delta_2 \approx 0.43$ (peak 3) at $\text{AE} = 0.959$. After smoothing, peak 3 sits at +0.39 and peak 2 is only a shoulder. A fifth local maximum at +1.39 is the copy of peak 3 inside peak 4.

3 Why noise gives one peak: diffusive smearing

The binomial noise averages out as $N^{-1/2}$; the efficiency noise does not, because one p_n is shared by all molecules. Then $\ln N_n$ is a random walk with increments $\ln(1 + p_m)$ for every noisy cycle, and the first-passage time through $\ln T$ acquires a Gaussian spread

$$\begin{aligned} \text{Cq} &\simeq \text{Cq}_0 + \Delta_{\text{early}} + \eta, \\ s_\eta^2 &= n_{\text{noise}} c(\text{AE})^2 \sigma^2, \quad c(\text{AE}) = \frac{1}{(1 + \text{AE}) \ln(1 + \text{AE})}, \end{aligned} \quad (5)$$

with $c(0.959) = 0.759$ and $c(0.8) = 0.945$. The Cq histogram is the comb of early-time peaks convolved with a Gaussian of width s_η , and $\text{Var}(\text{Cq}) = \text{Var}_0 + s_\eta^2$. Table 1 checks this in the literal reading ($n_{\text{noise}} \simeq 39$, so $s_\eta = 4.75 \sigma$).

σ	s_η	predicted std	simulated std
0.02	0.095	0.286	0.287
0.04	0.190	0.330	0.327
0.06	0.285	0.393	0.380
0.10	0.475	0.547	0.499

Table 1: Variance additivity, literal reading, $\text{AE} = 0.959$, $\text{std}_0 = 0.270$. The simulated width falls below the prediction at large σ because clipping at $p = 1$ trims the noise.

A side peak of weight w at distance d from a dominant one survives as a separate maximum only if d/s_η exceeds 2.0 ($w = 0.5$), 3.3 ($w = 0.1$) or 3.6 ($w = 0.05$). At $\sigma = 0.1$, $s_\eta \approx 0.47$ exceeds every spacing, so only one peak is left.

4 Literal reading: results

The shape of panel (a) matches the paper; the absolute position is 0.83 cycles late because $T = 2^{38}$ instead of $2^{37.2}$. Panels (b,c) are far broader than the published panel (b): the deterministic estimate $s_\eta = 0.47$ explains the single peak.

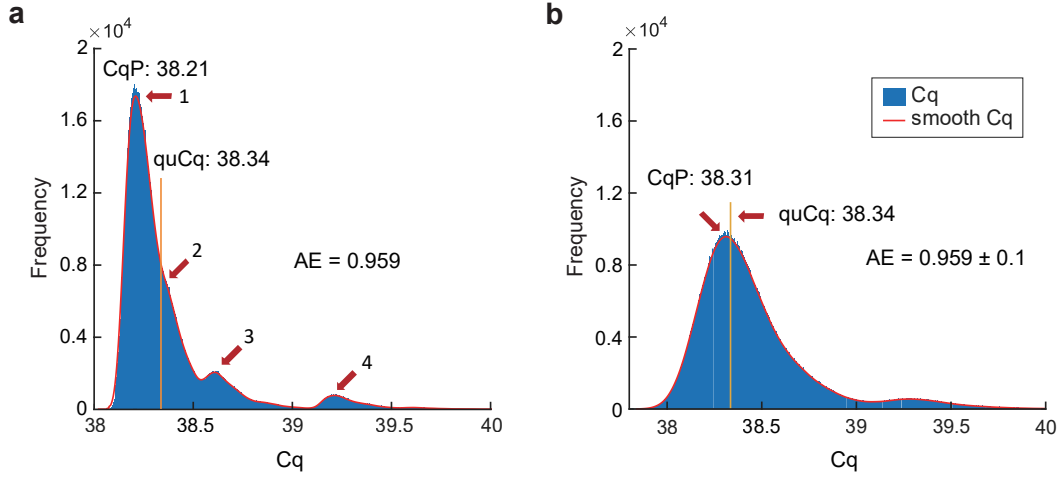


Figure 1: Published Fig. S3 of the quPCR supplement, for comparison: (a) $AE = 0.959$, four peaks, $CqP = 38.21$; (b) $AE = 0.959 \pm 0.1$, two broad peaks, $CqP = 38.31$.

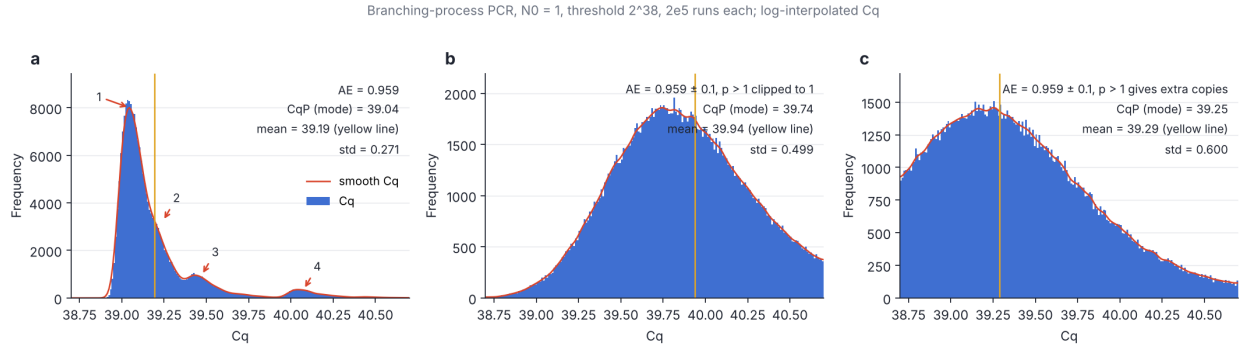


Figure 2: Literal reading, $AE = 0.959$, 2×10^5 runs per panel. (a) Fixed AE: four peaks, main peak at 39.04. (b) $\sigma = 0.1$ every cycle, p clipped at 1: one broad peak, std 0.50, shifted late by 0.7 because $\langle \min(p, 1) \rangle = 0.936$. (c) Same, but $p > 1$ gives extra copies: no shift, same width.

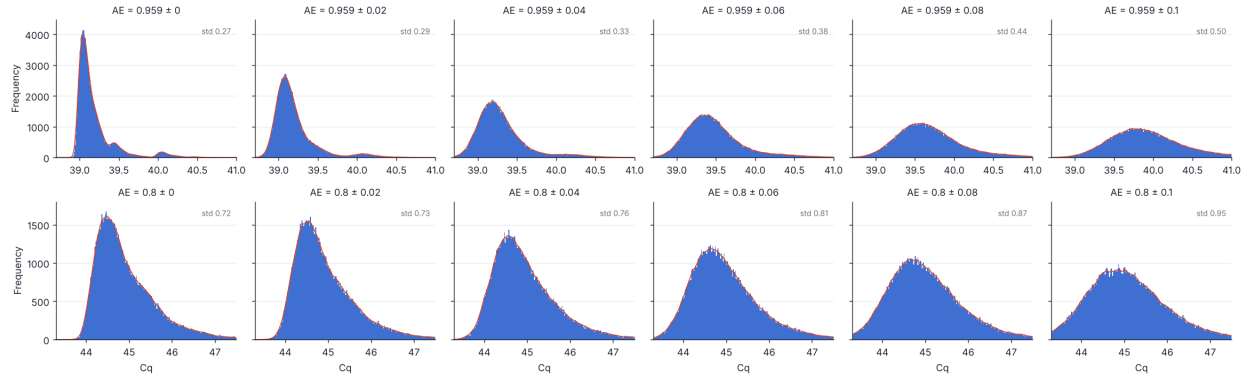


Figure 3: Literal reading, σ swept from 0 to 0.1, 10^5 runs per panel. Top $AE = 0.959$, bottom $AE = 0.8$. The full sweep in steps of 0.0025 is [figures/cq.sigma.sweep.gif](#) and [.mp4](#).

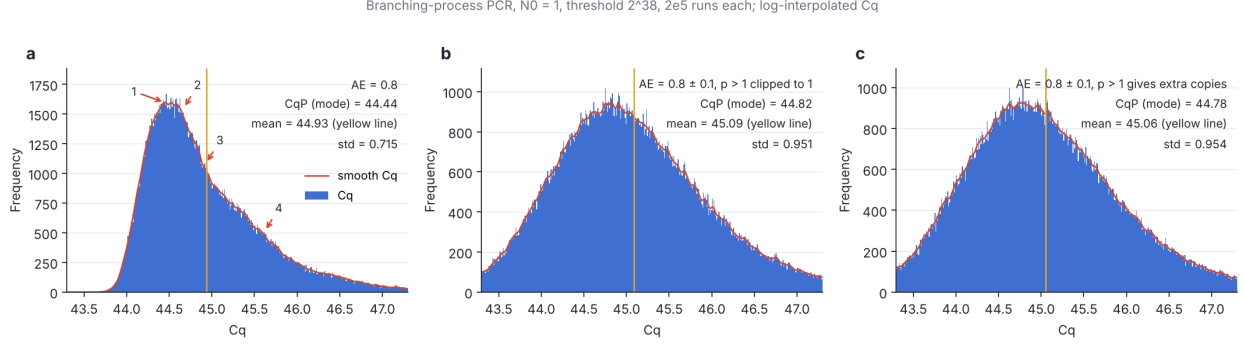


Figure 4: Literal reading, $AE = 0.8$. The side peaks are shoulders already at $\sigma = 0$ (failure probability 20% in cycle 1), and the intrinsic width (std 0.72) exceeds the noise contribution at $\sigma = 0.1$ (total std 0.95).

5 Authors' protocol: exact reproduction

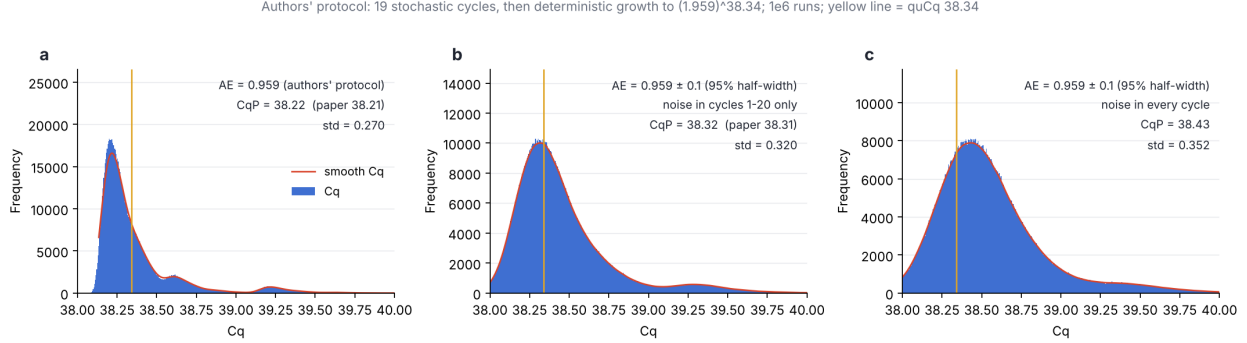


Figure 5: Authors' protocol, 10^6 runs, binning and smoothing as in their scripts. (a) $AE = 0.959$; (b) $AE = 0.9593$, $\sigma = 0.051$ in cycles 1–20; (c) the same noise in every cycle, which the authors' code omits.

	CqP (sim.)	CqP (paper)	std
$AE = 0.959$	38.22	38.21	0.270
$AE = 0.959 \pm 0.1$ (95%), noise in cycles 1–20	38.32	38.31	0.320
same, noise in every cycle	38.43	—	0.352

Table 2: Reproduction of Fig. S3. With $n_{\text{noise}} = 19$ and $\sigma = 0.051$, Eq. (5) gives $s_\eta = 0.17$, which leaves peak 4 standing.

6 Noise scan: when do the four peaks return?

We scan $\sigma \in [0, 0.05]$ in steps of 0.0025 in the authors' protocol, 5×10^5 runs per point, with the authors' experimental smoothing (Gaussian window of 0.15 cycles). Both AE values use the same physical threshold $T = 1.959^{38.34}$ copies, so $quCq(0.8) = 43.86$. The visibility of a side peak is $V = (\max - \text{dip})/\max$, set to zero when there is no local maximum.

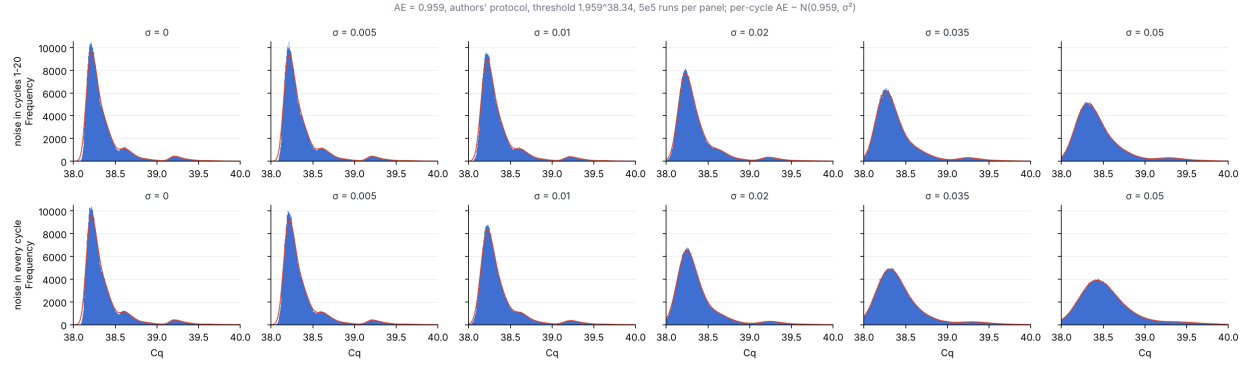


Figure 6: AE = 0.959, authors' protocol. Top: noise in cycles 1–20; bottom: noise in every cycle. The four-peak structure is back for $\sigma \lesssim 0.005$ –0.01.

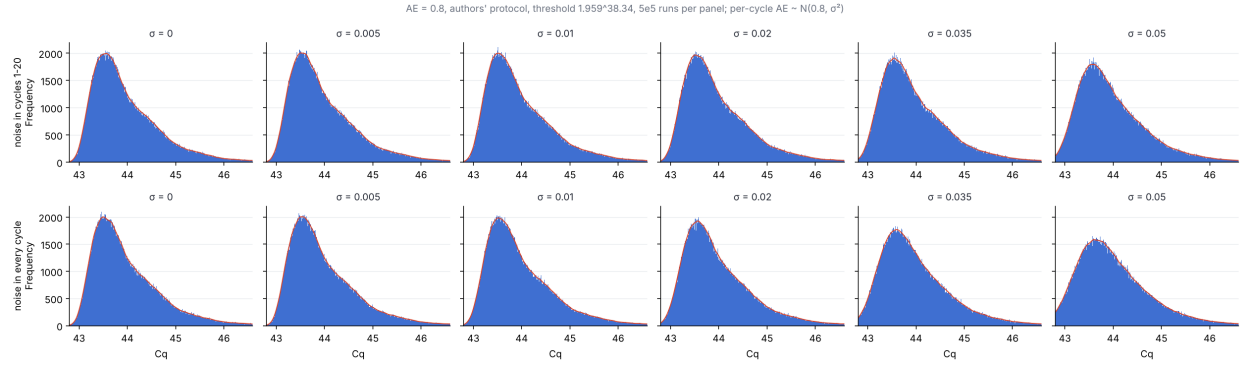


Figure 7: AE = 0.8, authors' protocol, same layout. Even at $\sigma = 0$ there is a single skewed maximum at $Cq = 43.56$ ($quCq = 43.86$); the side peaks at $+0.23$, $+0.49$ and $+1.00$ are shoulders, and the noise only widens the hump.

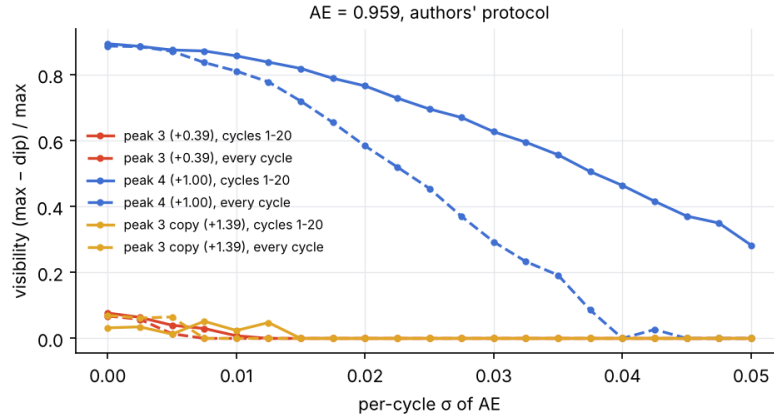


Figure 8: Side-peak visibility against σ . Solid: noise in cycles 1–20 (authors); dashed: noise in every cycle.

AE	side peak (offset)	noise in cycles 1–20	noise in every cycle
0.959	peak 3 (+0.39)	$\sigma_c \approx 0.011$	$\sigma_c \approx 0.006$
0.959	peak 4 (+1.00)	> 0.05	≈ 0.04
0.8	peaks 2–4	no separate maxima, even at $\sigma = 0$	

Table 3: Largest σ at which each side peak is still a separate maximum. $\sigma = 0$ visibilities at AE = 0.959: peak 3 0.08, peak 4 0.89.

AE = 0.8: no peaks to return to. Early failures are five times more likely ($1 - \text{AE} = 20\%$ in cycle 1, and one of two molecules fails in cycle 2 with probability $2\text{AE}(1 - \text{AE}) = 32\%$), so the teeth of the comb carry comparable weights at spacings 0.23 and 0.49, below their own widths. The distribution is a single skewed hump at every σ (Fig. 7), and σ enters only through the width. Eq. (5) with $c(0.8) = 0.945$ predicts std 0.746 ($\sigma = 0.05$, $n_{\text{noise}} = 19$) and 0.782 ($n_{\text{noise}} = 44$), against simulated 0.739 and 0.781 from $\text{std}_0 = 0.717$. Since std_0 is $2.7\times$ larger than at AE = 0.959 and there are no sharp features, the same σ is much harder to detect (Fig. 10).

7 What the experiment bounds

Peak 3 is not resolvable with ~ 300 wells. The authors’ data (`ExtremeDiluteR.xlsx`) contain 690 positive wells, of which 297 are single-copy ($38.07 \leq \text{Cq} \leq 39.5$). Bootstrapping N wells from the $\sigma = 0$ simulation, a local maximum near peak 3 appears in 86% of draws at $N = 300$, but also in 74% of draws from $\sigma = 0.02$, where no peak 3 exists. The two are separated only at $N \sim 10^4$ (99% against 0%). The peak-counting criterion $\sigma \lesssim 0.01$ is therefore not supported by the data.

Likelihood fit. For each σ the model density (5×10^5 runs) is convolved with Gaussian measurement noise, shifted by a free offset that absorbs quCq, truncated to the window and renormalised. Profiling the offset gives Table 4 and Fig. 9.

measurement sd	noise in cycles 1–20		noise in every cycle		$\Delta(-2 \ln L)$ at $\sigma = 0$
	best σ	95% bound	best σ	95% bound	
0	0.025	0.036	0.015	0.024	18.5 / 18.0
0.03	0.020	0.036	0.015	0.023	7.2 / 7.4
0.06	0.005	0.030	0.005	0.019	0.3 / 0.5

Table 4: Profile-likelihood fit of the per-cycle σ to 297 single-copy wells. 0.03 is the paper’s high-copy Cq SD. The S3B noise ($\sigma = 0.051$, cycles 1–20) has $\Delta(-2 \ln L) = 12.7\text{--}14.8$ and is excluded by the authors’ own data.

The upper bound is robust to the measurement-noise assumption; whether $\sigma > 0$ is preferred is not. Any well-to-well broadening (threshold or baseline variation, a static per-well spread of AE) is degenerate with σ in the width, so the preference for $\sigma \approx 0.015\text{--}0.02$ at measurement sd 0.03 is not evidence for per-cycle fluctuations specifically.

Wells needed. If the truth is $\sigma = 0$, the expected likelihood-ratio statistic against σ with N wells is $2N \text{KL}(p_0 \| p_\sigma)$, with the shift profiled out. Setting it to 3.84 gives the number of single-copy wells needed to exclude σ at 95% (Fig. 10). For AE = 0.959 and 297 wells the expected bound is

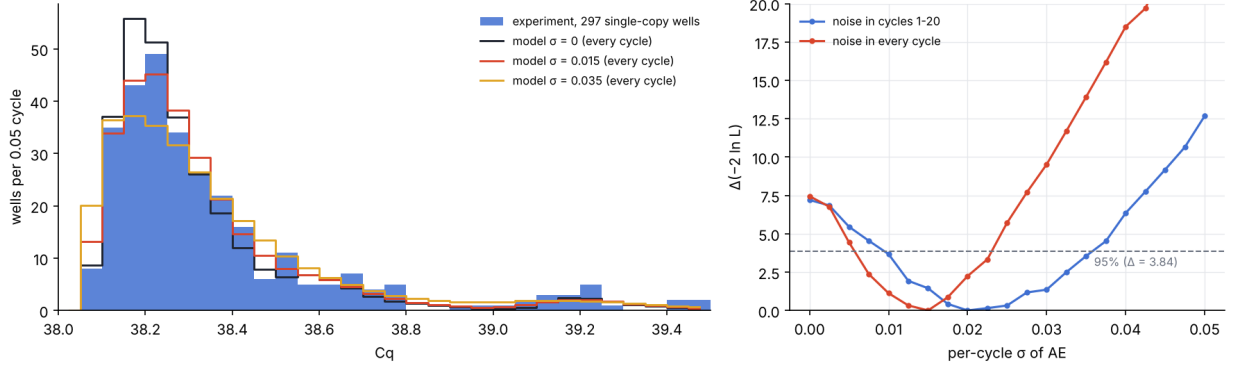


Figure 9: Left: experimental single-copy Cq histogram against the model at $\sigma = 0$, the best fit $\sigma = 0.015$ and the excluded value $\sigma = 0.035$ (noise in every cycle, measurement sd 0.03). Right: profile likelihood.

$\sigma \approx 0.012$ (every cycle) to 0.017 (cycles 1–20). The observed bounds are looser, 0.023 and 0.036, because the data are slightly wider than the $\sigma = 0$ model.

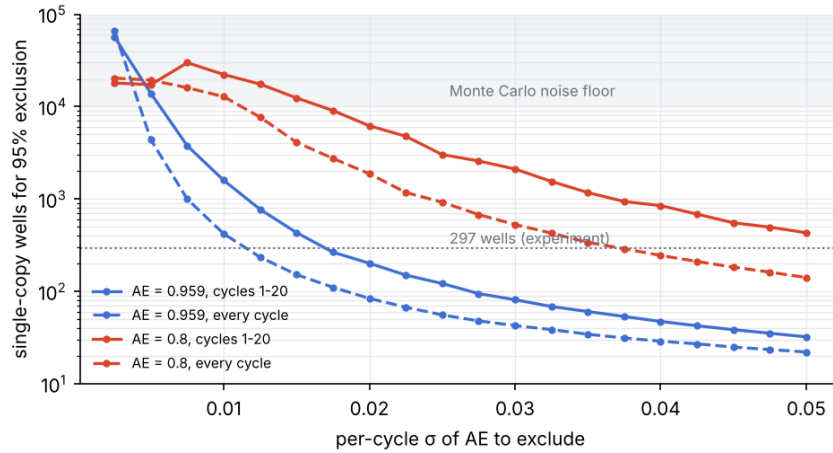


Figure 10: Single-copy wells needed to exclude a per-cycle σ at 95% when the truth is $\sigma = 0$ (measurement sd 0.03). Values above $\sim 10^4$ are limited by the Monte Carlo noise of the 5×10^5 -run histograms.

8 Summary: how accurate is AE?

1. **The mean of AE barely enters the peak pattern.** Peak 4 is at +1 cycle for any AE, and $\Delta_{2,3}$ change by $c(\text{AE}) \delta \text{AE} \approx 0.5\%$ for the quoted $\delta \text{AE} = 0.006$. The mean is set by the dilution series, not by the peaks.
2. **The peaks bound the per-cycle, per-well fluctuation of AE.** The four-peak structure returns for $\sigma \lesssim 0.006$ –0.011 (peak 3) and peak 4 survives to $\sigma \approx 0.04$. With the authors' 297 single-copy wells the actual 95% bound is $\sigma \lesssim 0.02$ –0.036, i.e. AE within roughly ± 0.04 –0.07 of 0.959 cycle to cycle. Fig. S3B's $\sigma = 0.051$ is excluded, but “no physically plausible fluctuation” is not shown.

3. **Fluctuations common to a plate are unconstrained.** The simulations draw noise independently per well. Thermocycler drift shared by all wells of a plate shifts the histogram without broadening it.
4. $AE = 0.8$. There are no four peaks to return to: the side peaks are shoulders already at $\sigma = 0$. Noise only adds width to an intrinsic std of 0.72, so excluding a given σ takes 10–30 $\times$ more single-copy wells than at $AE = 0.959$. With 297 wells the expected 95% bound is $\sigma \approx 0.036$ (noise in every cycle) or above 0.05 (cycles 1–20), against 0.012–0.017 at $AE = 0.959$.